scalar QED

QFT #7 KNU ko yurang

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Charged particle

If ϕ has a charge Q, ϕ is transformed

$$\phi \to e^{iQ\alpha}\phi$$

complex conjugate field

$$\phi^* \to e^{-iQ\alpha}\phi^*$$

If α is local, $\alpha = \alpha(x)$, to make invariant the Lagrangian, we introduced covariant derivative

$$D_\mu = \partial_\mu - ieQA_\mu$$

Where A_μ is transformed as $A_\mu \to A_\mu + \frac{1}{e}\partial_\mu\alpha$,

Charged particle

We already know A_μ is massless because of gauge invariance. Our Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \left|D_\mu\phi\right|^2 - m^2|\phi|^2$$
$$= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \left|\partial_\mu\phi\right|^2 - m^2|\phi|^2 + ieQA_\mu[\phi^*\partial^\mu\phi - (\partial^\mu\phi^*)\phi] + e^2Q^2A_\mu^2|\phi|^2$$

ϕ and ϕ^* Equation of motion

$$(\Box + m^2)\phi = i(-eQA^\mu)\partial_\mu\phi + i\partial_\mu(-eQA^\mu\phi) + \left(-eQA_\mu\right)^2\phi$$
$$(\Box + m^2)\phi^* = i(eQA^\mu)\partial_\mu\phi^* + i\partial_\mu(eQA^\mu\phi) + \left(eQA_\mu\right)^2\phi^*$$

Quantizing complex scalar fields

So, ϕ and ϕ^* couple to A_μ with opposite charge with same mass. in a second-quantized relativistic theory, the radiation process $\phi \rightarrow \gamma\phi$, automatically implies that $\gamma \rightarrow \phi\phi^*$ is also possible.

⇒ We must be able to produce these ϕ^* particles. In other words, in a relativistic theory with a massless spin-1 field, antiparticles must exist.

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left(a_p e^{-ipx} + b_p^\dagger e^{ipx}\right)$$

$$\phi^*(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left(a_p^\dagger e^{ipx} + b_p e^{-ipx}\right)$$

Quantizing complex scalar fields

We can conclude b_p annihilates particle of opposite charge and the same mass.

> Matter coupled to massless spin-1 particles automatically implies the existence of antiparticles, which are particles of identical mass and opposite charge.

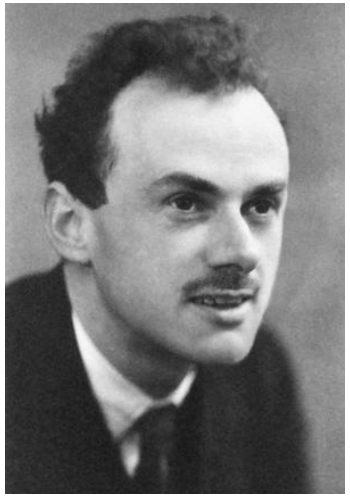
※ Historical Note : holes

From the Dirac equation, there are two energy eigen value

$$E = \pm\sqrt{\vec{p}^2 + m^2}$$

Positive one and negative one

Quantizing complex scalar fields



$a^{\dagger}e^{ipx} + c^{\dagger}e^{-ipx}$

Dirac interpret negative energy part as “particle with negative energy”

But we know, if particle has negative energy, $[H, c^{\dagger}] = -Ec^{\dagger}$,system is unstable

Why wouldn’t Dirac know something even we know? So Dirac devised the Dirac sea : He interpreted that the universe is already filled with negative-energy electrons. Because of Pauli exclusion principle, energy can’t drop more.

Suppose an electron is removed from the Dirac sea. The interpretation is that a positively charged particle, called a "hole," is then created in that empty space within the Dirac sea

Quantizing complex scalar fields

Did the boson collapse or something? (in Kwak Cheolyong mengki)

Nowadays QFT, we interpret negative energy part as antiparticle, $c^\dagger = b$.

Not creation of negative energy but annihilation of antiparticle. We can interpret $b^\dagger$ automatically creation of antiparticle.

Of course, in semiconductor field, concept of hole is used in absence of electron of valence band.

Feynman rules for scalar QED

For $Q = -1$ the Lagrangian is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \left|\partial_\mu\phi\right|^2 - m^2|\phi|^2 - ieA_\mu[\phi^*\partial^\mu\phi - (\partial^\mu\phi^*)\phi] + e^2A_\mu^2|\phi|^2$$

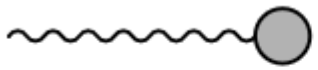


$$= \frac{i}{p^2 - m^2 + i\varepsilon}.$$

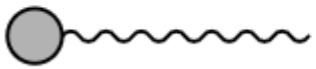


$$= \frac{-i}{p^2 + i\varepsilon}\left[g_{\mu\nu} - (1-\xi)\frac{p_\mu p_\nu}{p^2}\right],$$

propagator



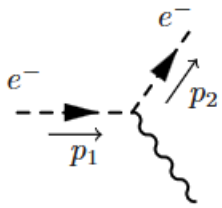
$$= \epsilon_\mu(p) \quad \text{(incoming)},$$



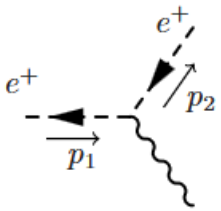
$$= \epsilon_\mu^\star(p) \quad \text{(outgoing)}.$$

For external photon leg. Interaction $-ieA_\mu[\phi^*\partial^\mu\phi - (\partial^\mu\phi^*)\phi]$ gives $\phi^*\phi A$ vertex,

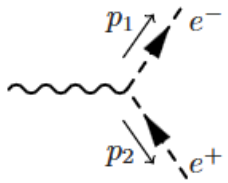
Feynman rules for scalar QED



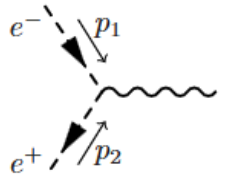
$= ie(-p_\mu^1 - p_\mu^2).$



$= ie(p_\mu^1 + p_\mu^2).$

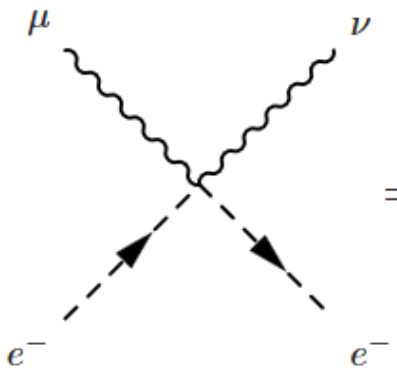


$= ie(-p_\mu^1 + p_\mu^2).$



$= ie(-p_\mu^1 + p_\mu^2).$

You can get easily. And from interaction $+e^2 A_\mu^2 |\phi|^2$, we can get seagull vertex (seagull : 갈매기)

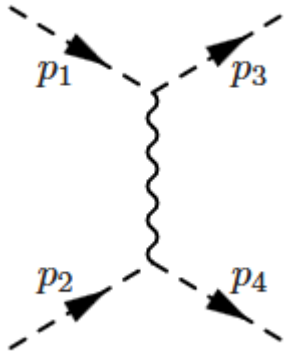


$= 2ie^2 g_{\mu\nu}.$

2 comes from symmetric factor of A.

Scattering in scalar QED

Moller scattering, $e^-e^- \rightarrow e^-e^-$, there are two diagrams, t channel and u channel.

$$i\mathcal{M}_t = $$ 

$$= (-ie)^2\left(p_1^\mu + p_3^\mu\right)\frac{-i\left[g_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{t}\right]}{t}\left(p_2^\nu + p_4^\nu\right)$$

where $k^\mu = \left(p_1^\mu - p_3^\mu\right)$, and $t = k^2$.

Here, $k_\mu\left(p_1^\mu + p_3^\mu\right) = p_1^2 - p_3^2 = 0$ So, gauge term in the propagator vanish.

Scattering in scalar QED

$$i\mathcal{M}_t = (-ie)^2\left(p_1^\mu + p_3^\mu\right)\frac{-ig_{\mu\nu}}{t}\left(p_2^\nu + p_4^\nu\right)$$

So,

$$\mathcal{M}_t = e^2\frac{(p_1+p_3)(p_2+p_4)}{t} = e^2\frac{p_1p_2 + p_1p_4 + p_3p_2 + p_3p_4}{t}$$

While Mandelstam variable relation is

$$p_1p_2 = \frac{s - m_1^2 - m_2^2}{2}, \qquad p_3p_4 = \frac{s - m_3^2 - m_4^2}{2}$$

$$p_1p_3 = \frac{m_1^2 + m_3^2 - t}{2}, \qquad p_2p_4 = \frac{m_2^2 + m_4^2 - t}{2}$$

$$p_1p_4 = \frac{m_1^2 + m_4^2 - u}{2}, \qquad p_2p_3 = \frac{m_2^2 + m_3^2 - u}{2}$$

Scattering in scalar QED

$$p_1 p_2 = \frac{s}{2} - m^2 = p_3 p_4$$
$$p_1 p_3 = m^2 - \frac{t}{2} = p_2 p_4$$
$$p_1 p_4 = m^2 - \frac{u}{2} = p_2 p_3$$

So,

$$\mathcal{M}_t = e^2 \frac{s-u}{t}$$

Note, u channel is just p_3 and p_4 change of t channel, $s \to s, t \to u, u \to t$,

$$\mathcal{M}_u = e^2 \frac{s-t}{u}$$

Scattering in scalar QED

So,

$$\left(\frac{d\sigma}{d\Omega}\right)_{\mathrm{CM}} = \frac{1}{64\pi^2 s}|\mathcal{M}|^2$$
$$= \frac{e^2}{64\pi^2 s}\left|\frac{s-u}{t}+\frac{s-t}{u}\right|^2$$

Note that, this is scalar QED, there is no relative minus sign.

Ward identity and gauge invariance

A general matrix element involving an internal photon will be $M^{\mu\nu}\Pi_{\mu\nu}$.

Where

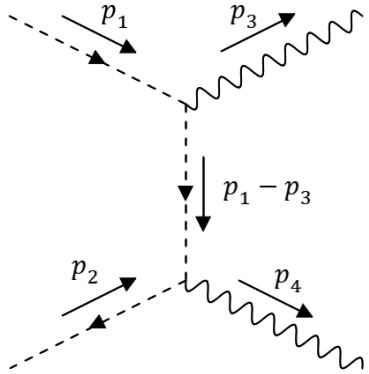
$$i\Pi_{\mu\nu} = \frac{-i\left[g_{\mu\nu} - (1-\xi)\frac{p_\mu p_\nu}{p^2}\right]}{p^2 + i\epsilon}$$

To make our physics independent of ξ, must satisfy $M^{\mu\nu}p_\mu p_\nu = 0$. This looks similar Ward identity $p_\mu M^\mu = 0$ where $\mathcal{M} = \epsilon_\mu M^\mu$: longitudinal mode of photon is unphysical.

Let’s check Ward identity first

Ward identity and gauge invariance

For example, $e^- e^+ \to \gamma\gamma$.

$i\mathcal{M}_t =$ 

$= (-ie)^2 \epsilon_\mu^*(p_3)\epsilon_\nu^*(p_4)\left(2p_1^\mu - p_3^\mu\right)\frac{i}{t-m^2}(p_4^\nu - 2p_2^\nu)$

$$M_t^{\mu\nu} = -e^2 \frac{2p_1^\mu p_4^\nu - 4p_1^\mu p_2^\nu - p_3^\mu p_4^\nu + 2p_3^\mu p_2^\nu}{p_3^2 - 2p_1 p_3}$$

u channel is $p_3 \leftrightarrow p_4$(or $p_1 \leftrightarrow p_2$) so,

$$M_u^{\mu\nu} = -e^2 \frac{2p_2^\mu p_4^\nu - 4p_2^\mu p_1^\nu - p_3^\mu p_4^\nu + 2p_3^\mu p_1^\nu}{p_3^2 - 2p_2 p_3}$$

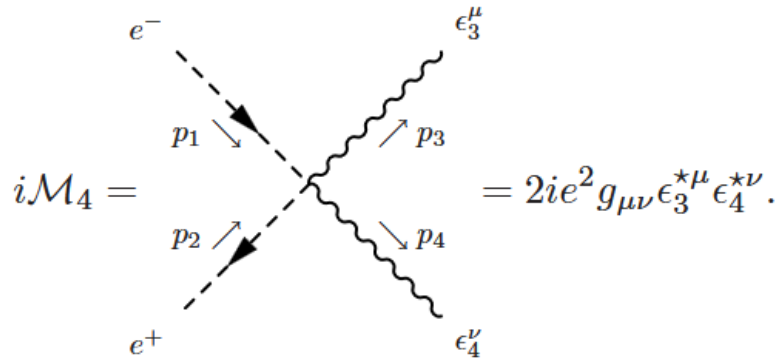
Ward identity and gauge invariance

Let's check Ward identity. Here, we didn't assume photon is on-shell and transverse polarization yet!

$$p_{3\mu}\left(M_t^{\mu\nu}+M_u^{\mu\nu}\right)=-p_{3\mu}\left(e^2\frac{2p_1^\mu p_4^\nu-4p_1^\mu p_2^\nu-p_3^\mu p_4^\nu+2p_3^\mu p_2^\nu}{p_3^2-2p_1p_3}+e^2\frac{2p_2^\mu p_4^\nu-4p_2^\mu p_1^\nu-p_3^\mu p_4^\nu+2p_3^\mu p_1^\nu}{p_3^2-2p_2p_3}\right)$$
$$=2e^2(p_4^\nu-p_1^\nu-p_2^\nu)\neq 0$$

The resolution is the missing diagram involving the 4-point vertex!

– Seagull one!



$$i\mathcal{M}_4 = = 2ie^2 g_{\mu\nu}\epsilon_3^{\star\mu}\epsilon_4^{\star\nu}.$$

$$M^{\mu\nu}_{\text{seagull}}=2e^2g^{\mu\nu}$$

Ward identity and gauge invariance

Finally we can check

$$p_{3\mu}\left(M_t^{\mu\nu} + M_u^{\mu\nu}\right) + p_{3\mu}M_{\text{seagull}}^{\mu\nu} = 2e^2(p_4^\nu - p_1^\nu - p_2^\nu) + 2e^2 p_3^\nu = 0$$

the **Ward identity would be satisfied even if the external photon states were not physical**; for example, if they were in a loop. In fact, that is exactly what we need for gauge invariance, so the same calculation can be used to prove ξ independence :

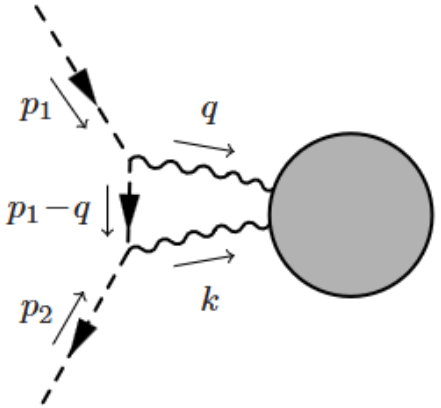
The Ward identity is not gauge invariance itself, but it is the manifestation of gauge invariance at the level of scattering amplitudes.

And this example shows that the full amplitude is gauge invariant, but not each individual term.

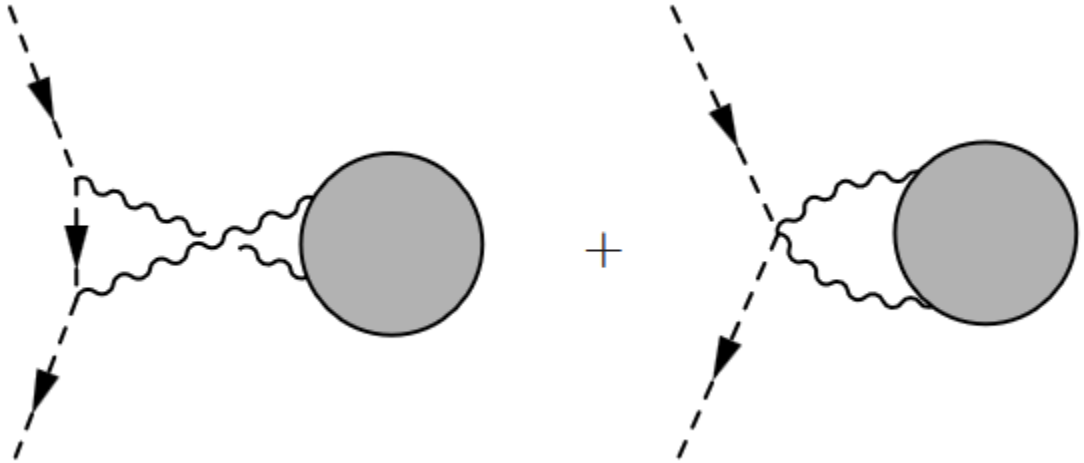
Ward identity and gauge invariance

$$i\mathcal{M}_t = e^2 \int \frac{d^4q}{(2\pi)^4}\frac{d^4k}{(2\pi)^4}\delta^4(p_1+p_2-q-k)\frac{(q^\mu - 2p_1^\mu)(k^\nu - 2p_2^\nu)}{q^2-2qp_1}$$
$$\times \Pi_{\mu\alpha}(q)\Pi_{\nu\beta}(k)X^{\alpha\beta}(q,k)$$

$i\mathcal{M}_t =$



to see invariance, we need to evaluate another two diagrams



Ward identity and gauge invariance

$$\mathcal{M}_t + \mathcal{M}_u + \mathcal{M}_{\text{seagull}} = e^2 \int \frac{d^4q}{(2\pi)^4}\frac{d^4k}{(2\pi)^4}\delta^4(p_1 + p_2 - q - k)$$
$$\times \left[\frac{(q^\mu - 2p_1^\mu)(k^\nu - 2p_2^\nu)}{q^2 - 2qp_1} + \frac{(q^\mu - 2p_2^\mu)(k^\nu - 2p_1^\nu)}{q^2 - 2qp_2} + 2g^{\mu\nu}\right]\Pi_{\mu\alpha}(q)\Pi_{\nu\beta}(k)X^{\alpha\beta}(q,k)$$

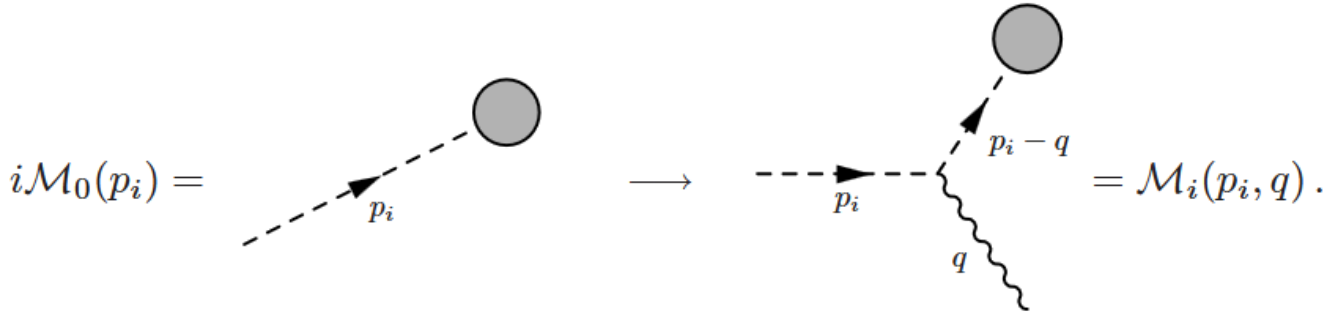
Since ξ term of $\Pi_{\mu\alpha}(q)$ is $\xi q_\mu q_\alpha$ so,

$$(\xi \text{ term}) = 2\xi e^2 \int \frac{d^4q}{(2\pi)^4}\frac{d^4k}{(2\pi)^4}\delta^4(p_1 + p_2 - q - k)(k^\nu + q^\nu - p_1^\nu - p_2^\nu)q_\alpha\Pi_{\nu\beta}(k)X^{\alpha\beta}(q,k)$$
$$= 0$$

We will study gauge invariance and Ward identity more deeply at chapter 14, path integral(Ward-Takahashi identity)

Charge conservation

Imagine we have some diagram with lots of external legs and loops and things. Say the matrix element for this process is M_0. Now tack on an outgoing photon of momentum q_μ and polarization ϵ_μ onto an external leg.



Amplitude is

$$\mathcal{M}_i(p_i, q) = (-iQe)\left(p_i^\mu + \left(p_i^\mu - q^\mu\right)\right)\frac{i}{(p_i - q)^2 - m^2}\epsilon_\mu(q)\mathcal{M}_0(p_i - q)$$

$$= -Qe\frac{p_i \cdot \epsilon}{p_i \cdot q}\mathcal{M}_0(p_i - q) \approx -Qe\frac{p_i \cdot \epsilon}{p_i \cdot q}\mathcal{M}_0(p_i) \ \text{(soft limit)}$$

Charge conservation

For incoming e^+, $\mathcal{M}_i(p_i, q) \approx +Qe\frac{p_i \cdot \epsilon}{p_i \cdot q}\mathcal{M}_0(p_i)$

We have to consider all external leg to put soft photon. If we had many different particles with different charges,

$$\mathcal{M}(p_i, \epsilon) \approx e\mathcal{M}_0\left[\sum_{\text{incoming}} Q_i \frac{p_i \cdot \epsilon}{p_i \cdot q} - \sum_{\text{outgoing}} Q_i \frac{p_i \cdot \epsilon}{p_i \cdot q}\right]$$

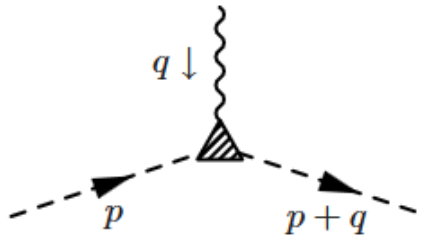
It must be invariant under Lorentz transform, $\mathcal{M}(p_i, \epsilon) = \mathcal{M}(p_i', \epsilon')$. We already learned,

$\epsilon \to \epsilon' = \epsilon + \lambda q$ for a little group transformation of q s.t. $q \to q' = \Lambda q = q$. Under this transf,

$$0 = \delta\mathcal{M} = e\mathcal{M}_0\left[\sum_{\text{incoming}} Q_i - \sum_{\text{outgoing}} Q_i\right] (!!!)$$

Charge conservation

Is it for only scalar QED? Nope. For example,

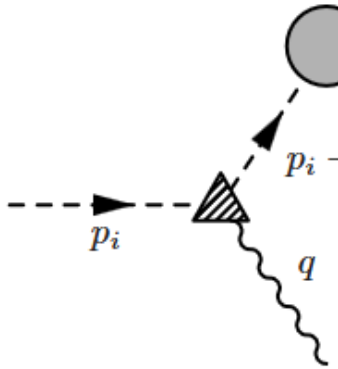


$$= -ie\Gamma_\mu(p,q)\,.$$

μ must contract with polarization. Since available 4 vectors are q and p. So,

$$\Gamma^\mu = 2p^\mu F(p^2, q^2, p\cdot q) + q^\mu G(p^2, q^2, p\cdot q)$$

F and G called form factor. From $q^\mu \epsilon_\mu = 0$, we can discard G. And $p^2 = m^2, q^2 = 0$, by dimensional analysis, $F = F\left(\frac{p\cdot q}{m^2}\right)$. i denoted particles and $F(0)$ for $q \to 0$ soft limit. So,



$$\approx -eF_i(0)\frac{p_i\cdot\epsilon}{p_i\cdot q}\mathcal{M}_0(p_i)\,.$$

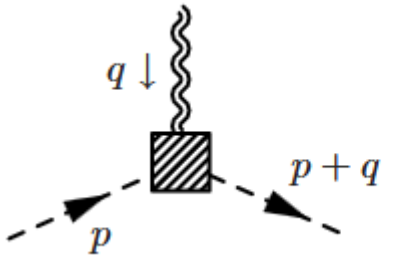
Charge conservation

$$0 = \delta\mathcal{M} = -e\mathcal{M}_0 \left[\sum_{\text{incoming}} F_i(0) - \sum_{\text{outgoing}} F_i(0) \right]$$

Thus, we get the same result as before, and moreover produce a general definition of the charge $Q_i = -F_i(0)$.

Massless spin 1 particles imply conservation of charge. Note, $\frac{p\cdot\epsilon}{p\cdot q}$ factor (called eikonal factor) is *spin independent universal form*.

spin 2 and higher



$= -i\Gamma^{\mu\nu}(p,q)$

Massless spin 2 field $h_{\mu\nu}$. It has polarization tensor $\epsilon_{\mu\nu}$ with DoF 2.

$\Gamma^{\mu\nu}$ must constract polarization. available 4 momentum is p, q

So

$$\Gamma^{\mu\nu} = 2p^{\mu}q^{\nu}F\left(\frac{p\cdot q}{m^2}\right)$$

Then eikonal-like factor must be $\frac{p^{\mu}q^{\nu}\epsilon_{\mu\nu}}{p\cdot q}$Therefore modified amplitude is

$$\mathcal{M} = \mathcal{M}_0\left[\sum_{\text{incoming}} F_i(0)\frac{p_i^{\mu}q^{\nu}\epsilon_{\mu\nu}}{p_i\cdot q} - \sum_{\text{outgoing}} F_i(0)\frac{p_i^{\mu}q^{\nu}\epsilon_{\mu\nu}}{p_i\cdot q}\right]$$

spin 2 and higher

Under little group transformation,

$$\epsilon_{\mu\nu} \to \epsilon_{\mu\nu} + \xi_\mu q_\nu + \xi_\nu q_\mu + \xi q_\mu q_\nu$$

Just replace $\epsilon_{\mu\nu} \to \xi_\nu q_\mu$ in $\mathcal{M}$

$$0 = \delta\mathcal{M} = \mathcal{M}_0 \xi_\nu \left[\sum_{\text{incoming}} \kappa_i \, p_i^\nu - \sum_{\text{outgoing}} \kappa_i p_i^\nu \right]$$

Where $F_i(0) = \kappa_i$. From 4 momentum conservation, $\kappa_i = \kappa$ for any i.

> But that is exactly what gravity does! All particles gravitate with the same strength, $\kappa_i \equiv \frac{1}{M_{\rm Pl}} = \sqrt{G_n}$. In other words, gravity is universal. So,
>
> Massless spin-2 particles imply gravity is universal.

spin 2 and higher

For massless spin 3

$$\sum_{\text{incoming}} \beta_i p_i^\mu p_i^\nu = \sum_{\text{outgoing}} \beta_i p_i^\mu p_i^\nu$$

$\mu = \nu = 0$ components :

$$\sum_{\text{incoming}} \beta_i E_i^2 = \sum_{\text{outgoing}} \beta_i E_i^2$$

that is, the sum of the squares of the energies times some charges are conserved. That is way too constraining. The only way out is if all the charges are 0, which is a boring, non-interacting theory of free massless spin-3 field. So,

There are no interacting theories of massless particles of spin greater than 2.

And in fact, no massless particles with spin > 2 have ever been seen. (Massive particles of spin > 2 are plentiful [Particle Data Group (Beringer *et al.*), 2012].)

For some fun…

$$p_1 = (E, 0,0, k), \qquad p_2 = (E, 0,0, -k)$$
$$p_3 = (E, k\sin\theta\,, 0, k\cos\theta), \qquad p_4 = (E, -k\sin\theta\,, 0, -k\cos\theta)$$

Let particles are all same species,

$$\sum_{\text{incoming}} \beta p^\mu p^\nu = \sum_{\text{outgoing}} \beta p^\mu p^\nu$$

$$2\beta \begin{pmatrix} E^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & k^2 \end{pmatrix} = 2\beta \begin{pmatrix} E^2 & 0 & 0 & 0 \\ 0 & k^2\sin^2\theta & 0 & k^2\cos\theta\sin\theta \\ 0 & 0 & 0 & 0 \\ 0 & k^2\cos\theta\sin\theta & 0 & k^2\cos^2\theta \end{pmatrix}$$

Only $\cos\theta = \pm 1$ scattering allowed : If we let $\beta \neq 0$, scattering can occur only in the forward or backward direction.